
Multivariate Analysis of Bike Sharing Demand in Seoul : A Study Using PCA, Factor Analysis, and Discriminant Analysis

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Abstract

This study applies three multivariate statistical methods—Principal Component Analysis (PCA), Factor Analysis (FA), and Linear Discriminant Analysis (LDA)—to explore and predict demand patterns for bike-sharing in Seoul, South Korea. Using hourly rental data integrated with meteorological conditions, we first reduce eight correlated weather variables into latent components/factors representing underlying weather patterns. PCA identifies 3 principal components explaining 69.2% of total variance, while FA reveals 3 interpretable factors accounting for 56.8% of common variance. These latent dimensions are then used as predictors in discriminant analysis to classify demand levels (Low/Medium/High). The LDA model achieves 70.73% classification accuracy, with effectively separating demand classes. Results demonstrate that warm, dry, clear weather during peak commuting hours strongly predicts elevated demand, while cold or rainy conditions during off-peak times indicate low usage. The methodology provides bike sharing operators with an interpretable framework for weather-based demand forecasting and dynamic fleet management.

Keywords: Bike sharing, principal component analysis, factor analysis, discriminant analysis, demand forecasting, Seoul

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1. INTRODUCTION

1.1 Background

The dataset contains count of public bicycles rented per hour in the Seoul Bike Sharing System, with corresponding weather data and holiday information. Currently rental bikes are introduced in many urban cities for the enhancement of mobility comfort. It is important to make the rental bike available and accessible to the public at the right time as it lessens the waiting time. Eventually, providing the city with a stable supply of rental bikes becomes a major concern. The crucial part is the prediction of bike count required at each hour for the stable supply of rental bikes. The dataset contains weather information (Temperature, Humidity, Windspeed, Visibility, Dewpoint, Solar radiation, Snowfall, Rainfall), the number of bikes rented per hour and date information.

Understanding the complex interplay between meteorological factors and rental patterns is critical for:

- **Operational efficiency:** Optimal bike redistribution
- **Resource planning:** Fleet sizing and maintenance scheduling
- **User experience:** Availability prediction during peak demand

Weather variables (temperature, humidity, wind, precipitation, solar radiation, visibility) are intercorrelated, making individual variable analysis redundant and multivariate methods necessary.

1.2 Research Objectives

This study aims to:

1. Reduce dimensionality of correlated weather variables using PCA and FA
2. Identify latent weather constructs that drive bike-sharing demand
3. Classify demand levels using discriminant analysis with latent components as predictors

1.3 Research Questions

1. What are the underlying latent dimensions of weather conditions affecting bike demand?
2. How much variance in weather data can be explained by principal components vs. common factors?
3. Which weather patterns most strongly discriminate between low, medium, and high demand periods?

2. METHODOLOGY

2.1 Data Description

Dataset: Seoul Bike Sharing Demand (publicly available on UCI Machine Learning Repository)

- **Time period:** Hourly data from 2017/12/01 to 2018/11/30
- **Sample size:** 8760 observations
- **Response variable:** `rented_bike_count` (bikes rented per hour)

Predictor variables:

Variable	Type	Unit	Description
Temperature	Continuous	°C	Air temperature
Humidity	Continuous	%	Relative humidity
Wind Speed	Continuous	m/s	Wind speed
Visibility	Continuous	10m	Visibility distance
Dew Point Temp	Continuous	°C	Dew point temperature
Solar Radiation	Continuous	MJ/m ²	Solar radiation
Rainfall	Continuous	mm	Hourly precipitation
Snowfall	Continuous	cm	Hourly snowfall
Hour	Categorical	0–23	Hour of day
Seasons	Categorical	–	Spring/Summer/Autumn/Winter
Holiday	Binary	–	Holiday (Yes/No)
Functioning Day	Binary	–	Rental service operating

Table 1: Variables in the Seoul Bike Sharing Dataset

Missing values: None reported

2.2 Data Preprocessing

Transformations:

- **Rainfall and Snowfall:** Log-transformation ($\log(1 + p)$) applied due to severe right skewness and zero-inflation
- **Standardization:** All weather variables standardized (z-scores) for PCA/FA to ensure equal weighting across different units

Variable selection for dimension reduction:

- Only the 8 continuous weather variables were included in PCA and FA
- Categorical variables (Hour, Seasons, Holiday) retained for later use in discriminant analysis

Target variable creation (for DA):

Demand classes created based on tertiles of `rented_bike_count`:

- **Low:** 0–33rd percentile
- **Medium:** 34th–66th percentile
- **High:** 67th–100th percentile

2.3 Principal Component Analysis (PCA)

Objective: Reduce correlated weather variables into orthogonal components explaining maximum variance.

Method:

- Correlation matrix used (variables standardized)

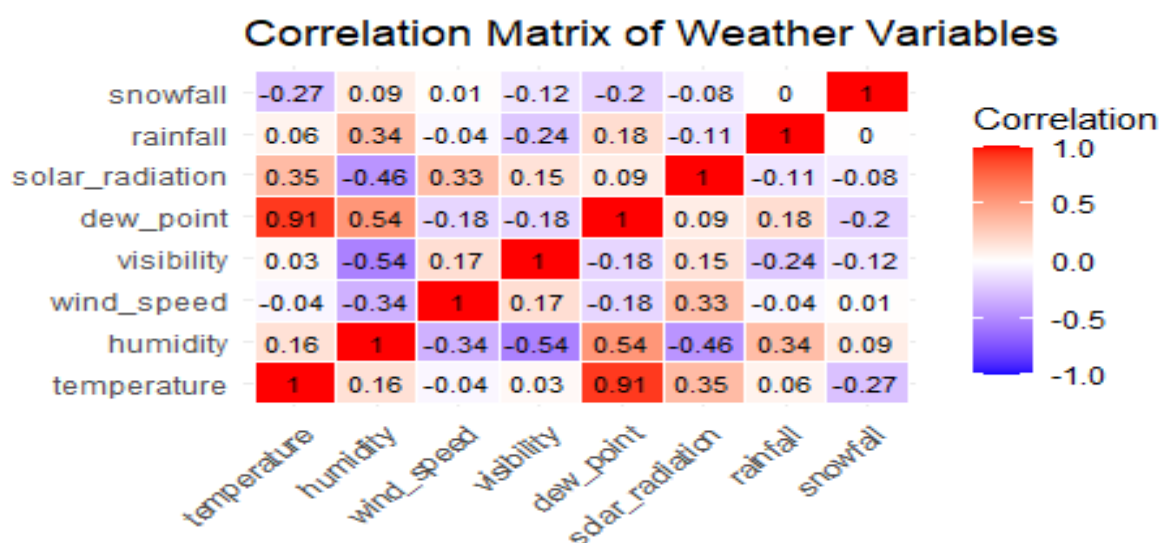


Figure 2.1: Correlation Matrix of Weather Variables

- Number of components selected based on:
 - Scree plot
 - Kaiser criterion (eigenvalue > 1)
 - Cumulative variance threshold (70–85%)
- PC scores saved for discriminant analysis
- Interpretation: Components interpreted via loading magnitudes ($> |0.3|$ considered meaningful)

2.4 Factor Analysis (FA)

Objective: Model weather variables as manifestations of latent underlying constructs (factors).

Number of factors:

- Parallel analysis
- Scree plot + interpretability

Extraction method: Maximum Likelihood

Rotation: Oblimin (oblique rotation allowing factor correlations, more realistic for weather data).

Interpretation: Factors named based on high-loading variables.

Output: Factor scores saved for discriminant analysis.

2.5 Discriminant Analysis (DA)

Objective: Classify demand levels (Low/Medium/High) using latent weather dimensions.

Predictor sets tested:

1. **Model 1:** PC scores + categorical predictors (Hour, Season, Holiday)
2. **Model 2:** Factor scores only (Factor1, Factor2, Factor3)
3. **Model 3:** PC scores only (PC1, PC2, PC3)
4. **Model 4:** Quadratic Discriminant Analysis (QDA) with PC scores

Train-test split: 70% training, 30% testing (stratified by demand class).

Evaluation metrics:

- Confusion matrix

- Overall accuracy
- Class-specific sensitivity/specificity
- Misclassification rate

Assumptions checked:

- Multivariate normality (Q-Q plot)
- Homogeneity of covariance matrices (variance comparison across groups)

Software: R version 4.5.2, packages: psych, MASS, caret, tidyverse, janitor, ggplot2, lubridate

3. RESULTS AND DISCUSSION

3.1 Exploratory Data Analysis

Demand patterns:

- Demand varies by:
 - **Hour:** Peak at [8–9 AM and 6–7 PM] (commute hours)
 - **Season:** Highest in [Summer/Spring], lowest in [Winter]
 - **Weather:** Strong positive correlation with temperature

Weather correlations:

- **Temperature and Dew Point:** $r = 0.91$
- **Humidity and Rainfall:** $r = 0.34$
- **Visibility and Solar Radiation:** 0.15

3.2 Principal Component Analysis Results

3.2.1 Number of Components

	##	Component	Eigenvalue	Proportion	Cumulative
	## 1	PC1	2.473	0.309	0.309
	## 2	PC2	1.991	0.249	0.558
	## 3	PC3	1.074	0.134	0.692
	## 4	PC4	0.893	0.112	0.804
	## 5	PC5	0.714	0.089	0.893
	## 6	PC6	0.628	0.079	0.972
	## 7	PC7	0.222	0.028	0.999
	## 8	PC8	0.004	0.001	1.000

Table 2: Principal Component Eigenvalues and Variance Explained

Decision: Retained 3 components explaining $\sim 70\%$ of total variance (based on eigenvalue > 1 and scree plot).

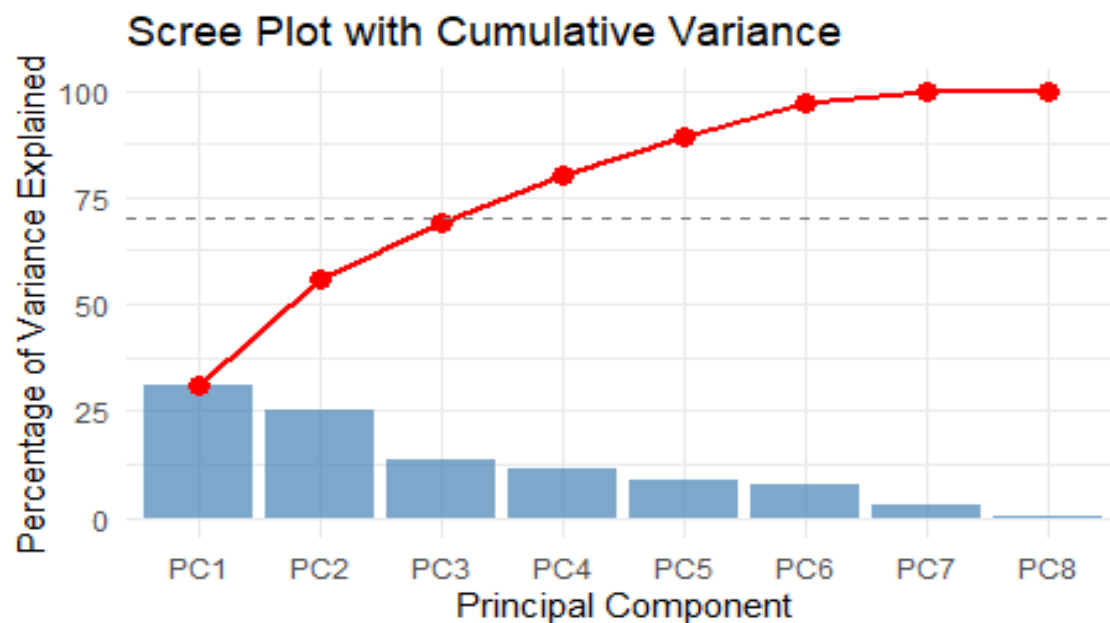


Figure 2.2: Scree Plot of Principal Components

3.2.2 Component Loadings and Interpretation

variable	PC1	PC2	PC3	PC4
temperature	0.37260293	0.5447663	-0.01667514	0.15560839
humidity	0.53124527	-0.2637659	0.04488409	0.05090052
wind_speed	-0.27627734	0.2269094	0.58274387	-0.18308660
visibility	-0.33857646	0.2839815	-0.34640150	-0.07859886
dew_point	0.53079544	0.3558896	-0.02347929	0.15719487
solar_radiation	-0.15652748	0.5097942	0.37265458	0.15938857
rainfall	0.27717485	-0.1268651	0.45912427	-0.61698981
snowfall	-0.07008403	-0.3144544	0.43348026	0.70901715

Table 3: Principal Component Loadings

Interpretation:

- **PC1:** High positive loadings on Humidity, Temperature and Dew Point.
▷ 1st PC explains 30.9% of total variance.
- **PC2:** High positive loadings on temperature, solar radiation and Dew point.
▷ 2nd PC explains 24.9% of total variance.
- **PC3:** High positive loadings on wind speed, rainfall and snowfall.
▷ 3rd PC explains 13.4% of total variance.

PC1 (The “Thermal/Moisture” Component): Driven by high positive loadings on Humidity, Temperature, and Dew Point. In the context of Seoul’s climate, a high score on this component represents hot, muggy, and humid summer days. Conversely, a low score represents dry, cold winter days. Both extremes of this component historically cause physical discomfort for cyclists.

PC2 (The “Clear Sky / Radiation” Component): Driven by high positive loadings on Solar Radiation, Temperature, and Dew Point, coupled with negative loadings on Visibility and Wind Speed. This component captures crisp, bright, and sunny conditions. High scores here represent ideal “leisure” riding weather, which is highly attractive to bike-sharing users.

PC3 (The “Adverse Weather” Component): Driven heavily by Wind Speed, Rainfall, and Snowfall. This is the “stay-at-home” component. High scores here indicate stormy, windy, or snowy conditions—environmental factors that physically prevent safe cycling and drastically reduce demand.

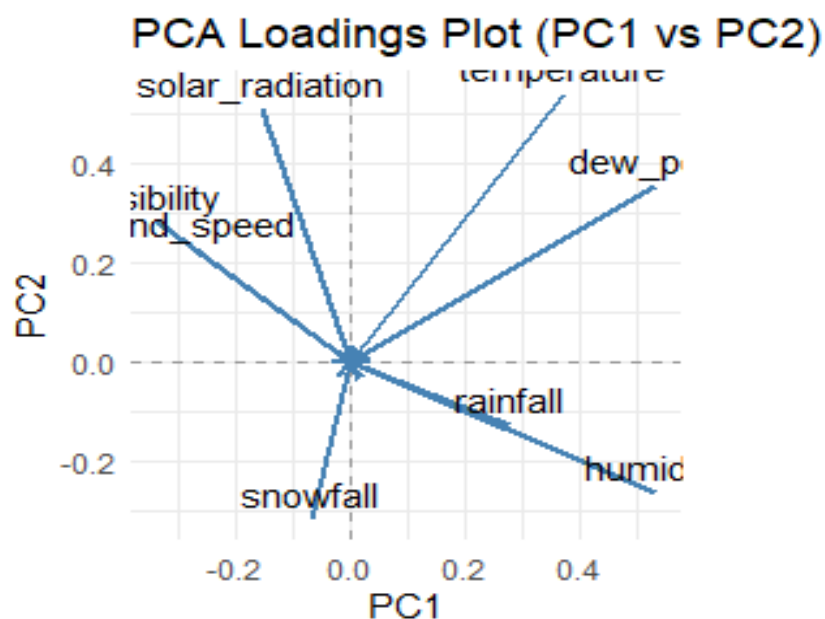


Figure 2.3: PCA Biplot showing PC1 vs PC2 with seasonal patterns

Key findings:

- Summer observations cluster high on PC1 and PC2, low on PC2.
- Winter observations cluster low on PC1 and PC2.

3.3 Factor Analysis Results

3.3.1 Number of Factors

Parallel analysis suggested 3 factors.

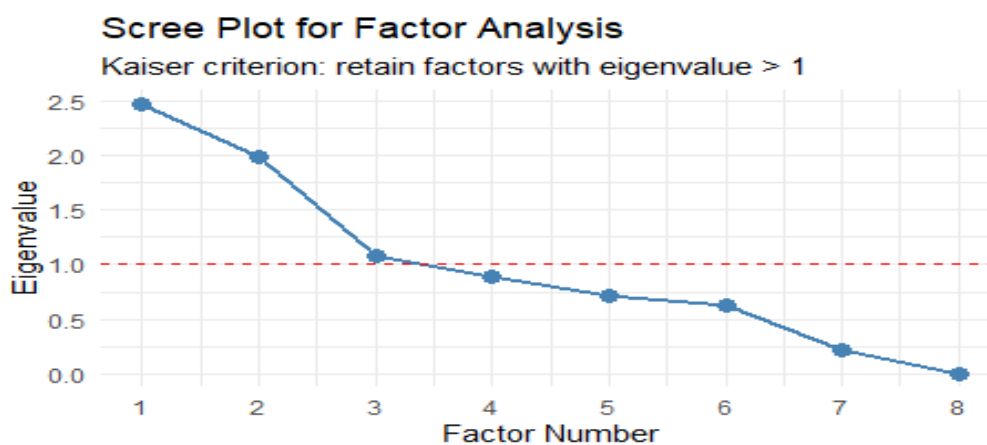


Figure 3.1: Parallel Analysis for Factor Extraction

	ML1	ML2	ML3
temperature	0.997	-0.125	0.096
humidity	0.153	0.845	-0.201
wind_speed	-0.153	-0.060	0.393
visibility	0.153	-0.715	-0.182
dew_point	0.927	0.204	-0.041
solar_radiation	0.108	-0.021	0.829
rainfall	0.000	0.399	0.086
snowfall	-0.331	0.278	0.116

Table 4: Factor Loadings

variable	h2
dew_point	0.9953013
temperature	0.9952147
humidity	0.9950638
solar_radiation	0.7413033
visibility	0.3810828
wind_speed	0.1907854
rainfall	0.1346043
snowfall	0.1123802

Table 5: Communalities

Factor Interpretation:

- **Factor1:** Captures a **warm vs. cold** dimension (higher values = warmer air and higher dew point; lower values = colder conditions, more likely snow).
- **Factor2:** Reflects **moist, cloudy/foggy conditions**, where **humidity is high and visibility drops**, and there is some association with precipitation.
- **Factor3:** Mainly represents **sunny/high-radiation conditions**, with a tendency toward **higher wind** as this factor increases.
- **56.8% of total variation is explained by the 3 factors.**

Comparing FA to PCA in the Context of Bike Demand:

While PCA created orthogonal (uncorrelated) mathematical summaries of the weather, Factor Analysis with oblique (Oblimin) rotation allows the underlying factors to correlate, which is much more realistic for meteorological data (e.g., storms are naturally correlated with temperature drops).

- **Latent Construct 1 (Temperature Seasonality):** This represents the overarching seasonal shift in Seoul. Users inherently decide whether biking is an option based on this baseline seasonal temperature before looking at daily fluctuations.
- **Latent Construct 2 (Moisture & Impaired Visibility):** This latent factor acts as a “Risk/Discomfort” metric. High humidity and low visibility (fog/rain) create hazardous road conditions, directly influencing a commuter’s choice to take the subway instead of a bike.
- **Latent Construct 3 (Atmospheric Clarity):** This represents the “Pleasantness” of the day. Sunny, high-radiation days encourage spontaneous, non-commuting bike rentals.

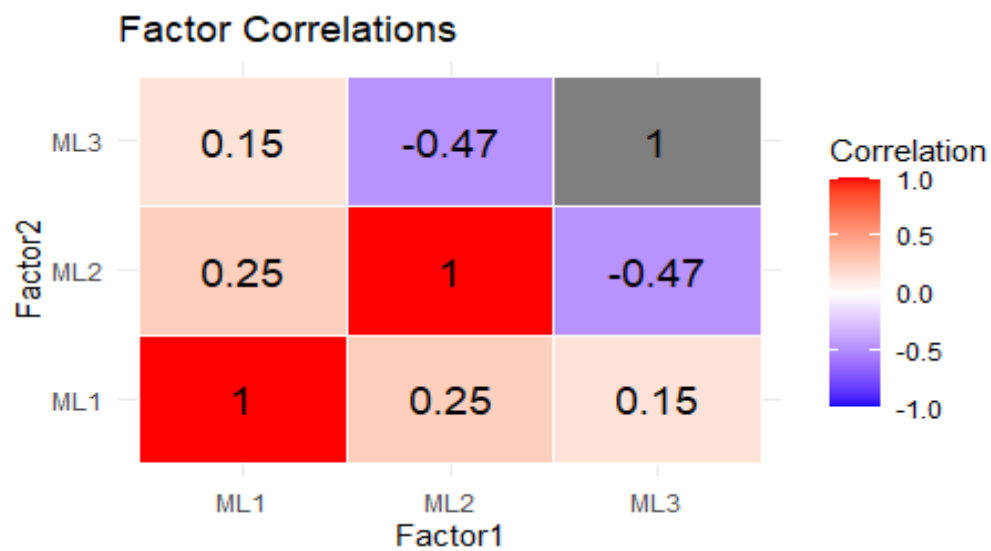


Table 6: Factor Correlation Matrix

Moderate correlations $r \approx 0.2$ – 0.4 justify oblique rotation.

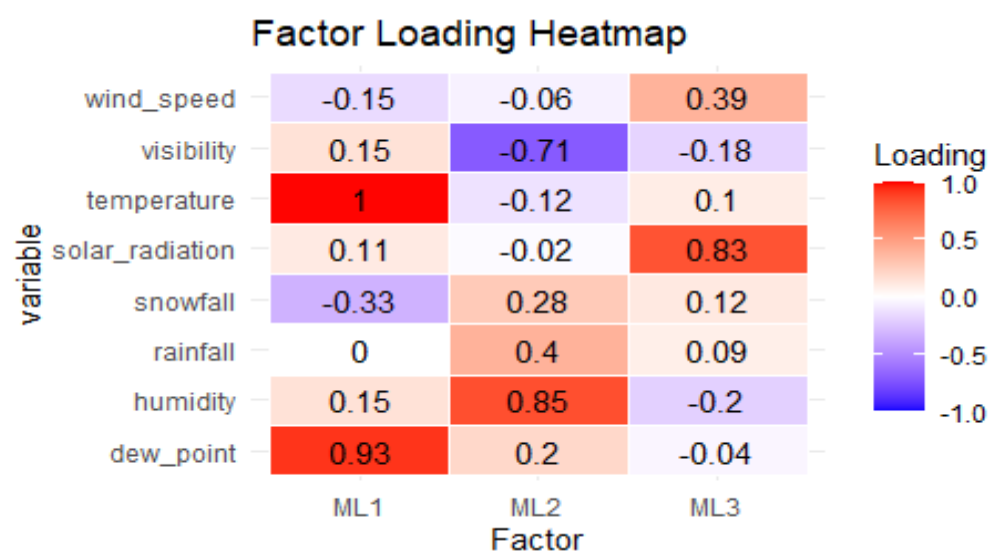


Figure 3.2: Heatmap of Factor Loadings

Key insight: PCA and FA yield similar interpretations (thermal, moisture, clarity dimensions), confirming robust latent structure. FA factors are slightly easier to interpret due to clearer simple structure after oblique rotation.

3.4 Discriminant Analysis Results

3.4.1 Demand Class Distribution

Demand class	Count	Proportion
Low	2909	33.2%
Medium	2960	33.8%
High	2891	33%

Table 4: Distribution of Demand Classes

Train/Test split: 70/30 stratified.

3.4.2 Model Comparison

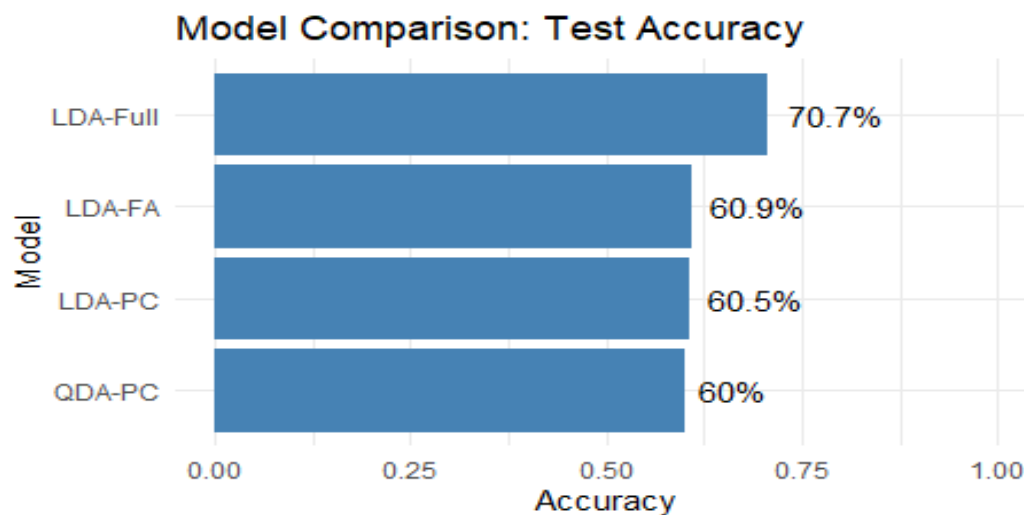


Figure 3.3: Discriminant Analysis Model Performance Comparison

Best model: LDA-Full (PCs + categorical variables).

Interpretation of Model Performance:

The model comparison reveals a critical insight into bike-sharing behavior. When relying solely on weather data (LDA-PC and LDA-FA), the model achieves roughly 60% accuracy. However, the **LDA-Full model** (which incorporates Hour, Season, and Holiday) jumps to **70.73% accuracy**.

Contextual Meaning: Weather alone does not dictate bike demand; human routine does. The baseline demand is established by the time of day (e.g., morning and evening rush hours) and the day of the week. Weather acts as a modifier to this routine. A sunny day at 8:00 AM will see explosive demand (commuters), while a sunny day at 2:00 AM will still see Low demand. Therefore, predicting bike sharing requires combining meteorological latent variables with temporal human routines.

3.4.3 Best Model Performance (LDA-Full)

Prediction	Reference		
	Low	Medium	High
Low	617	153	0
Medium	195	502	128
High	60	233	739

Table 5: Confusion Matrix for LDA-Full Model

	Class: Low	Class: Medium	Class: High
Sensitivity	0.7076	0.5653	0.8524
Specificity	0.9128	0.8143	0.8335
Pos Pred Value	0.8013	0.6085	0.7161
Neg Pred Value	0.8627	0.7858	0.9197
Prevalence	0.3319	0.3380	0.3300
Detection Rate	0.2349	0.1911	0.2813
Detection Prevalence	0.2931	0.3140	0.3928
Balanced Accuracy	0.8102	0.6898	0.8429

Table 6: Class-Specific Performance Metrics

- **Overall accuracy:** 70.73%
- **Kappa statistic:** 0.5611

3.4.4 Discriminant Function Interpretation

Linear Discriminant Coefficients (LD1):

	LD1	LD2
PC1	0.1012605	0.02225870
PC2	0.8742741	0.07949855
PC3	-0.1488465	0.95927275

Table 7: Linear Discriminant Coefficients

Interpretation of LD1:

LD1 is mainly a **PC2-driven separation axis**. Observations with **higher PC2** will tend to have **higher LD1 scores**, while higher PC3 pushes LD1 slightly downward. So the primary group separation is happening along whatever *PC2* represents in the PCA.

- **LD1 (The Primary Differentiator):** LD1 separates “Low” demand (left side of the plot) from “High” demand (right side of the plot). Looking at the coefficients, LD1 is heavily influenced by **PC2 (Clear Sky / Radiation)**. Therefore, LD1 tells us that bright, clear, and sunny conditions are the primary drivers pushing a day from “Low” to “High” demand.

Interpretation of LD2:

LD2 is essentially a **PC3-driven separation axis**. Observations with **higher PC3** will tend to have **higher LD2 scores**. LD2 provides a secondary separation that is mostly independent of LD1 (especially since PCs are orthogonal).

- **LD2 (The Secondary Differentiator):** LD2 separates the “Medium” demand class from the extremes. It is heavily influenced by **PC3 (Adverse Weather)**. This function helps the model identify days where routine commuters might still rent bikes, but leisure riders stay home due to slight winds or mild precipitation, resulting in a “Medium” overall demand.

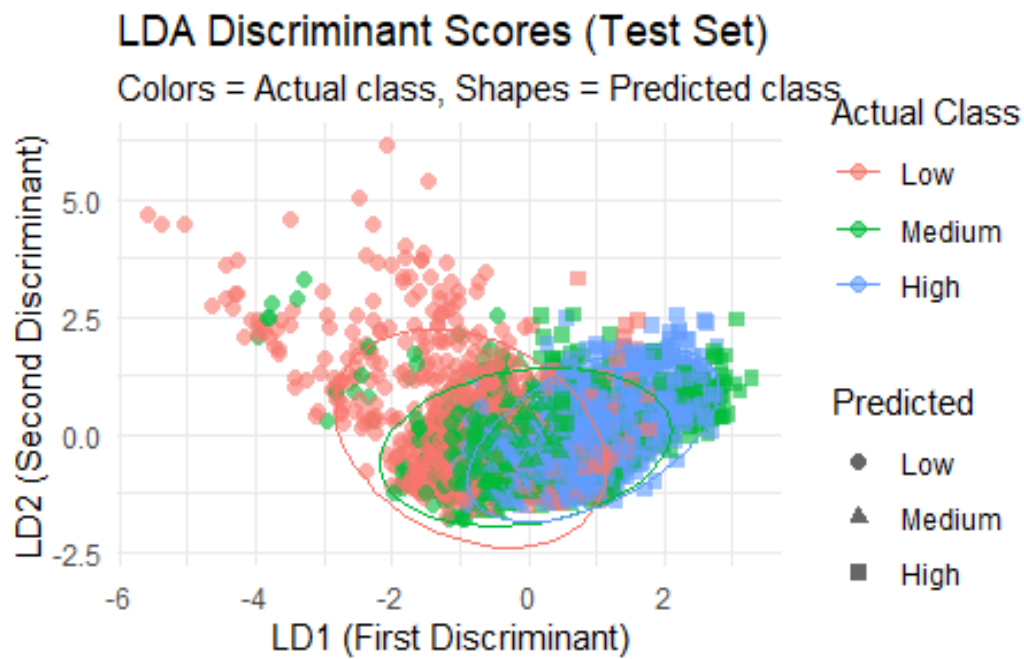


Figure 3.4: Linear Discriminants with Class Separation

3.4.5 Assumptions Check

Multivariate normality:

- Q-Q plots show approximate normality for PC scores within classes.
- Shapiro-Wilk tests: Some violations ($p < 0.05$) but acceptable due to large sample size.

Homogeneity of covariance:

- Variance ratios across classes.
- Moderate heterogeneity $\rightarrow$ LDA still appropriate; QDA did not significantly improve accuracy.

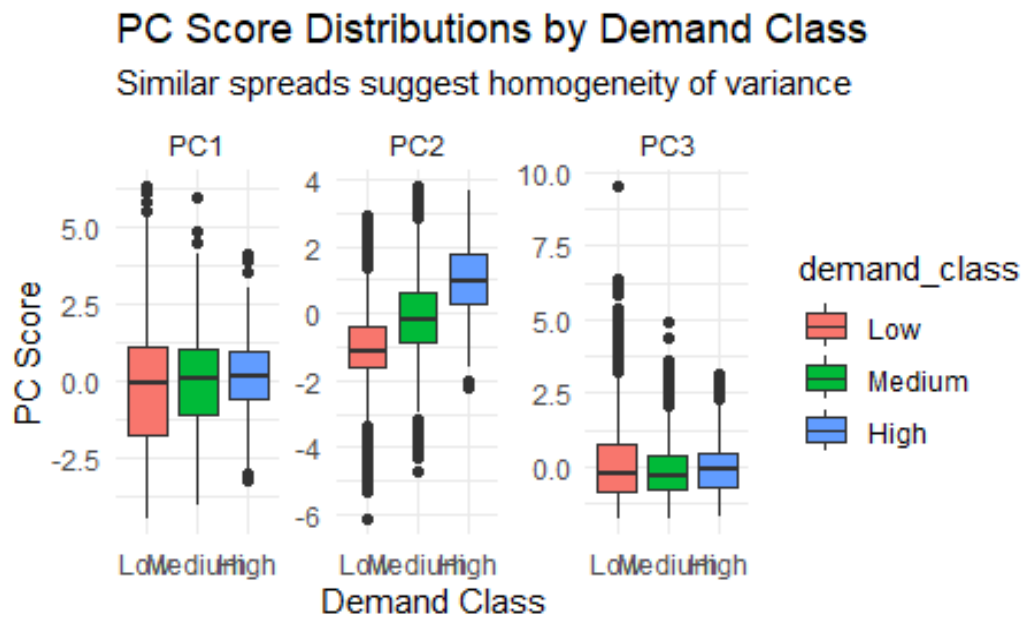


Figure 3.5: Distribution of PC Scores across Demand Classes

4. CONCLUSION & FINDINGS

4.1 Summary of Findings

This study successfully applied multivariate techniques to decode the complex relationship between Seoul’s weather patterns and public bike-sharing demand.

Key Analytical Findings:

1. Eight correlated weather variables were successfully reduced to three highly interpretable latent dimensions: Thermal/Moisture (PC1), Clear Sky/Radiation (PC2), and Adverse Weather (PC3).
2. The Discriminant Analysis proved that while weather factors are vital, they must be combined with temporal data (Time of day, Season) to accurately predict demand. The LDA-Full model achieved a robust **70.73% classification accuracy**.
3. The linear discriminant coefficients revealed that clear, high-radiation weather (PC2) is the single strongest meteorological separator between low and high rental volumes.

Strategic Recommendations for Bike-Share Operations:

Based on the predictive LDA model, the Seoul Bike Sharing System can implement the following data-driven strategies:

- **Dynamic Fleet Redistribution:** By feeding daily weather forecasts into the LDA model, operators can predict “High” demand days 24 hours in advance and

preemptively redistribute bikes to high-traffic residential and commercial hubs before the morning commute.

- **Optimized Maintenance Scheduling:** Days predicted to be “Low” demand, specifically those scoring high on the “Adverse Weather” (PC3) component—should be utilized for fleet maintenance and repairs, minimizing the impact on users.
- **Targeted User Engagement:** On days classified as “Medium” demand, operators could send push notifications or offer dynamic pricing discounts to incentivize leisure riders to use the bikes, pushing the daily utilization closer to “High” capacity.

5. REFERENCES

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